Last time: Schur-Weyl duality: As a $(GL_n \times S_k)$-repn,

$$V^{\otimes k} \cong \bigoplus_{\lambda \in Par(k,n)} V^\lambda \otimes S^\lambda.$$

Cor 75: In the highest wt. repn. $V^\lambda$, the wt. space multiplicities are given by the kostka numbers:

$$\dim \underbrace{V^\lambda_\mu}_{\substack{\text{wt. space for } \mu \\ \text{inside } V^\lambda}} = K_{\lambda\mu}.$$

Pf: Combining Lemma 73 and Thm. 74, treating $V^{\otimes k}$ as an $S_k$-repn, we have

$$V^{\otimes k} \cong \bigoplus_{\lambda \in Par(k,n)} V^\lambda \otimes S^\lambda \cong \bigoplus_{\substack{\lambda,\mu \\ \lambda \trianglerighteq \mu}} (V_\mu \cap (V^\lambda \otimes S^\lambda))$$

$$\cong \bigoplus_{\substack{\lambda,\mu \\ \lambda \trianglerighteq \mu}} (V^\lambda_\mu \otimes S^\lambda))$$

and on $V_\mu$, this gives

$$\bigoplus_\lambda K_{\lambda\mu} S^\lambda \cong M^\mu \cong V_\mu \cong \bigoplus_\lambda (V^\lambda_\mu \otimes S^\lambda) \cong \bigoplus_\lambda (\dim V^\lambda_\mu) S^\lambda$$

$\square$

# Symmetric functions

Def 76: The ring of symmetric functions is the ring $\Lambda = \mathbb{C}[x_1, x_2, \dots]^{S_n}$, where the $S_n$ action permutes the variables. Setting $x_{n+1} = x_{n+2} = \dots = 0$ gives sym. polys. in $n$ vars. Several important bases, all homog. of deg $|\lambda|$.

- elementary symmetric functions

$$e_k = \sum_{i_1 < \dots < i_k} x_{i_1} \cdots x_{i_k} \qquad e_\lambda = e_{\lambda_1} e_{\lambda_2} \cdots$$

- complete homogeneous symmetric functions

$$h_k = \sum_{i_1 \le \dots \le i_k} x_{i_1} \cdots x_{i_k} \qquad h_\lambda = h_{\lambda_1} h_{\lambda_1} \cdots$$

- monomial symmetric functions

$$m_\lambda = \sum_{\substack{i_1, \dots, i_\ell \\ \text{distinct}}} x_{i_1}^{\lambda_1} \cdots x_{i_\ell}^{\lambda_\ell}$$

- power sum symmetric functions

$$p_k = \sum_{i \ge 1} x_i^k \qquad p_\lambda = p_{\lambda_1} p_{\lambda_2} \cdots$$

- Schur functions (basis)

$$s_\lambda = \sum_{\substack{\text{SSYT } T \\ \text{of shape } \lambda}} x^{wt(T)} = \sum_{\mu : \text{comp.}} K_{\lambda\mu} x^\mu = \sum_{\substack{\mu : \text{part} \\ \mu \trianglelefteq \lambda}} K_{\lambda\mu} m_\mu$$

$$s_{(k)} = h_k, \quad s_{(1^k)} = e_k$$

Some Sym. fun. identities (see e.g. [Macdonald, Ch 1])

- Weyl character:

$$s_\lambda = \frac{\sum_{w \in S_n} (-1)^w x^{w(\lambda+\rho)}}{\sum_{w \in S_n} (-1)^w x^{w\rho}}$$

In particular, it is the character of the highest wt repn $V^\lambda$

- Cauchy identity:

$$\prod_{i,j} \frac{1}{1-x_i y_j} = \sum_\lambda s_\lambda(x) s_\lambda(y) = \sum_\mu \frac{1}{z_\mu} p_\mu(x) p_\mu(y)$$

see HW3 #2

- Pieri rule:

$$h_k s_\mu = \sum_\lambda s_\lambda,$$

where the sum is over all $\lambda$ s.t. $|\lambda| = |\mu| + k$ and $\lambda'_i - \mu'_i \leq 1 \ \forall i$.

- Jacobi-Trudi rule:

$$s_\lambda = \det[h_{\lambda_i + j - i}]_{1 \leq i,j \leq \ell(\lambda)} = \det[e_{\lambda'_i + j - i}]_{1 \leq i,j \leq \ell(\lambda')}$$

- Newton identities:

$$h_n = \frac{1}{n} \sum_{r=1}^{n} p_r h_{n-r} \quad \text{and} \quad e_n = \frac{1}{n} \sum_{r=1}^{n} (-1)^{r-1} p_r e_{n-r}$$

Prop 77: Let $g \in GL_n$ have evalues $x_1, \dots, x_n$, and let $w \in S_k$ have cycle type $\mu = (\mu_1, \dots, \mu_\ell)$. Then,

$$\mathrm{Tr}_{V^{\otimes k}}(g, w) = p_\mu(x_1, \dots, x_n)$$

Pf: By continuity and conjugation, we can assume $g = \mathrm{diag}(x_1, \dots, x_n)$. Then,

$$(g, w) \cdot e_I = x_{i_1} \cdots x_{i_k} e_{w \cdot I}$$

where $w \cdot I = (i_{w^{-1}(1)}, \dots, i_{w^{-1}(k)})$.

Now, $e_{w \cdot I} = e_I$ iff $i_j$ is constant on each cycle of $w$. For such an $I$, let $j_1, \dots, j_\ell$ denote the values of $i$ on each conj. class.

Thus we have

$$\begin{aligned}
\mathrm{Tr}_{V^{\otimes k}}(g, w) &= \sum_{\substack{I \\ w \cdot I = I}} x_{i_1} \cdots x_{i_k} \\
&= \sum_{1 \le j_1, \dots, j_\ell \le n} x_{j_1}^{\mu_1} \cdots x_{j_\ell}^{\mu_\ell} \\
&= p_{\mu_1}(x) p_{\mu_2}(x) \cdots p_{\mu_\ell}(x) \\
&= p_\mu(x).
\end{aligned}$$

$\square$

Let $R^k$ be the space of class functions on $S_k$, $R = \bigoplus_{k \geq 0} R^k$

Let $\Lambda^k$ be the space of homog. sym. funs. of deg $k$.

Recall: $\chi^\lambda$: character of $S^\lambda$, $w_\mu$: perm. of cycle type $\mu$

Def 78: The Frobenius characteristic map $ch: R \to \Lambda$ is the map

$$ch(f) = \sum_{\mu \vdash k} \frac{1}{z_\mu} f_\mu P_\mu ,$$

where $f_\mu$ is the value of $f$ on $w_\mu$

Thm 79:

a) $ch(\chi^\lambda) = s_\lambda$

b) $ch$ is a bijective linear map

c) $ch$ is an isometry w.r.t. the std. inner prod. on class functions and the inner prod. on $\Lambda^k$ defined by $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda\mu}$

d) $ch$ is an algebra isomorphism w.r.t. mult. in $\Lambda$ and the induction product

$$R^k \times R^m \to R^{k+m}$$

$$\chi \cdot \psi \mapsto \mathrm{Ind}_{S_k \times S_m}^{S_{k+m}} (\chi \otimes \psi)$$

Pf: a) Consider the $S_k$ class function $w \mapsto \mathrm{Tr}_{V^{\otimes k}}(g,w)$.

By Prop 77, $\mathrm{Tr}_{V^{\otimes k}}(g,w) = P_\mu(x)$, where $x_1,\dots,x_n$ are the evalues of $g$.

On the other hand, by Schur-Weyl duality, if $n \geq k$,

$$\mathrm{Tr}_{V^{\otimes k}}(g,w) = \sum_{\lambda \vdash k} \underbrace{\mathrm{Tr}_{V^\lambda}(g)}_{s_\lambda(x)} \cdot \chi^\lambda(w).$$

Equating, and applying the Frobenius characteristic map gives

$$\sum_{\mu \vdash k} \frac{1}{z_\mu} P_\mu(x) P_\mu(y) = \sum_{\lambda \vdash n} s_\lambda(x)\, \mathrm{ch}(\chi^\lambda)(y).$$

By the Cauchy identity and linear independence of the $s_\lambda(y)$, we have $\mathrm{ch}(\chi^\lambda) = s_\lambda$.

b), c) Follow since the $\chi^\lambda$ and $s_\lambda$ are orthonormal bases

d) Frobenius reciprocity (HW 5?)

$\square$

Cor 80 (Frobenius character formula):

$$s_\lambda = \sum_{\mu \vdash k} \frac{1}{z_\mu} \chi^\lambda(w_\mu) P_\mu,$$

so $z_\mu^{-1} \chi^\lambda(w_\mu)$ is the coeff. of $P_\mu$ in the expansion of $s_\lambda$ in the power sum basis.

Pf: Apply the def'n of the Frobenius characteristic and Thm 79(a). $\square$